

```
In [3]: !pip install tensorflow
```

Requirement already satisfied: tensorflow in c:\users\hp\anaconda3\lib\site-packages (2.21.0)
Requirement already satisfied: absl-py>=1.0.0 in c:\users\hp\anaconda3\lib\site-packages (from tensorflow) (2.5.0)
Requirement already satisfied: astunparse>=1.6.0 in c:\users\hp\anaconda3\lib\site-packages (from tensorflow) (1.6.3)
Requirement already satisfied: flatbuffers>=25.9.23 in c:\users\hp\anaconda3\lib\site-packages (from tensorflow) (25.12.19)
Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in c:\users\hp\anaconda3\lib\site-packages (from tensorflow) (0.7.0)
Requirement already satisfied: google_pasta>=0.1.1 in c:\users\hp\anaconda3\lib\site-packages (from tensorflow) (0.2.0)
Requirement already satisfied: libclang>=13.0.0 in c:\users\hp\anaconda3\lib\site-packages (from tensorflow) (18.1.1)
Requirement already satisfied: opt_einsum>=2.3.2 in c:\users\hp\anaconda3\lib\site-packages (from tensorflow) (3.4.0)
Requirement already satisfied: packaging in c:\users\hp\anaconda3\lib\site-packages (from tensorflow) (23.2)
Requirement already satisfied: protobuf<8.0.0,>=6.31.1 in c:\users\hp\anaconda3\lib\site-packages (from tensorflow) (7.35.1)
Requirement already satisfied: requests<3,>=2.21.0 in c:\users\hp\anaconda3\lib\site-packages (from tensorflow) (2.32.2)
Requirement already satisfied: setuptools in c:\users\hp\anaconda3\lib\site-packages (from tensorflow) (69.5.1)
Requirement already satisfied: six>=1.12.0 in c:\users\hp\anaconda3\lib\site-packages (from tensorflow) (1.16.0)
Requirement already satisfied: termcolor>=1.1.0 in c:\users\hp\anaconda3\lib\site-packages (from tensorflow) (3.3.0)
Requirement already satisfied: typing_extensions>=3.6.6 in c:\users\hp\anaconda3\lib\site-packages (from tensorflow) (4.16.0)
Requirement already satisfied: wrapt>=1.11.0 in c:\users\hp\anaconda3\lib\site-packages (from tensorflow) (1.14.1)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in c:\users\hp\anaconda3\lib\site-packages (from tensorflow) (1.81.1)
Requirement already satisfied: keras>=3.12.0 in c:\users\hp\anaconda3\lib\site-packages (from tensorflow) (3.15.0)
Requirement already satisfied: numpy>=1.26.0 in c:\users\hp\anaconda3\lib\site-packages (from tensorflow) (1.26.4)
Requirement already satisfied: h5py<3.15.0,>=3.11.0 in c:\users\hp\anaconda3\lib\site-packages (from tensorflow) (3.11.0)
Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in c:\users\hp\anaconda3\lib\site-packages (from tensorflow) (0.5.4)
Requirement already satisfied: wheel<1.0,>=0.23.0 in c:\users\hp\anaconda3\lib\site-packages (from astunparse>=1.6.0->tensorflow) (0.43.0)
Requirement already satisfied: rich in c:\users\hp\anaconda3\lib\site-packages (from keras>=3.12.0->tensorflow) (13.3.5)
Requirement already satisfied: namex in c:\users\hp\anaconda3\lib\site-packages (from keras>=3.12.0->tensorflow) (0.1.0)
Requirement already satisfied: optree in c:\users\hp\anaconda3\lib\site-packages (from keras>=3.12.0->tensorflow) (0.19.1)
Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\hp\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (2.0.4)
Requirement already satisfied: idna<4,>=2.5 in c:\users\hp\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (3.7)

Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\hp\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (2.2.2)

Requirement already satisfied: certifi>=2017.4.17 in c:\users\hp\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (2026.6.17)

Requirement already satisfied: markdown-it-py<3.0.0,>=2.2.0 in c:\users\hp\anaconda3\lib\site-packages (from rich->keras>=3.12.0->tensorflow) (2.2.0)

Requirement already satisfied: pygments<3.0.0,>=2.13.0 in c:\users\hp\appdata\roaming\python\python312\site-packages (from rich->keras>=3.12.0->tensorflow) (2.19.2)

Requirement already satisfied: mdurl~0.1 in c:\users\hp\anaconda3\lib\site-packages (from markdown-it-py<3.0.0,>=2.2.0->rich->keras>=3.12.0->tensorflow) (0.1.0)

```
In [5]: import tensorflow as tf
        from tensorflow.keras import layers, models
        import matplotlib.pyplot as plt
```

```
In [7]: mnist=tf.keras.datasets.mnist
        (train_images,train_labels),(test_images,test_labels)=mnist.load_data()
        train_images,test_images=train_images/255.0,test_images/255.0

        train_images=train_images.reshape((60000,28,28,1))
        test_images=test_images.reshape((10000,28,28,1))
```

```
In [9]: model=models.Sequential()
        model.add(layers.Conv2D(32,(3,3),activation='relu',input_shape=(28,28,1)))
        model.add(layers.MaxPooling2D((2,2)))

        model.add(layers.Conv2D(64,(3,3),activation='relu'))
        model.add(layers.MaxPooling2D((2,2)))

        model.add(layers.Flatten())
        model.add(layers.Dense(64,activation='relu'))
        model.add(layers.Dense(10,activation='softmax'))
        model.compile(optimizer='adam',loss='sparse_categorical_crossentropy',metrics=['accuracy'])
```

C:\Users\HP\anaconda3\Lib\site-packages\keras\src\layers\convolutional\base_conv.py:113: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

```
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuD NN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.

```
In [11]: model.fit(train_images,train_labels,epochs=5,validation_split=0.1)
         test_loss,test_acc=model.evaluate(test_images,test_labels)
         print('Test accuracy:',test_acc)
```

Epoch 1/5

1688/1688  **29s** 15ms/step - accuracy: 0.9491 - loss: 0.1672 - val_accuracy: 0.9852 - val_loss: 0.0571

Epoch 2/5

1688/1688  **25s** 15ms/step - accuracy: 0.9837 - loss: 0.0525 - val_accuracy: 0.9880 - val_loss: 0.0413

Epoch 3/5

1688/1688  **42s** 15ms/step - accuracy: 0.9885 - loss: 0.0361 - val_accuracy: 0.9893 - val_loss: 0.0382

Epoch 4/5

1688/1688  **40s** 15ms/step - accuracy: 0.9918 - loss: 0.0256 - val_accuracy: 0.9898 - val_loss: 0.0381

Epoch 5/5

1688/1688  **24s** 14ms/step - accuracy: 0.9932 - loss: 0.0209 - val_accuracy: 0.9902 - val_loss: 0.0407

313/313  **2s** 7ms/step - accuracy: 0.9888 - loss: 0.0341

Test accuracy: 0.9887999892234802